

Standard Model Particles from GF(27)* — Certificate Status of Quantum Numbers

FFGFT · J. Pascher 2026 · Doc. 346 / 347 / 348 / 349 / 353 (Overview: Doc. 356)

Structure of Doc. 356 (particle diagram): Each fermion is shown as a concentric ring diagram. Rings from inside out: **white inner circle** = symbol and charge Q ; to the left: Y_L (left-handed) and Y_R (right-handed); **colour ring** (quarks only, innermost after white) = $r/g/b$ colour charge **[B]** Doc. 346 — innermost because gluons confine colour; **middle ring** = chirality: left violet = S_d = left-handed, right orange = S_u = right-handed **[B]** Doc. 349; **outer ring** = Galois orbit colour (navy O1/O3, sky-blue O3, teal O2, berry O4) **[B]** Doc. 348; **red dashed box** = $SU(2)_L$ doublet pair; **black dashed circle** = Generation 3.

[B] Proved — algebraically established **[K] Confirmed** — numerically verified **[S] Speculative** — open **[X] Experimental** — established; FFGFT derivation open

Property	Galois origin / derivation	Doc. Status	Note
Electric charge Q	$Q \in \{0, -1, +2/3, -1/3\}$ from QR-bit + $N \bmod 3$	346/347 [B]	All 4 values without free parameters
Hypercharge Y	$Y = -1 + (4/3) \text{NQR}$; $\text{NQR} = (1 - (\frac{k}{13}))/2 \in \{0, 1\}$, so $Y = -1$ (QR) or $+1/3$ (NQR)	347 [S]	Theorem A; NQR-bit from GF(27)* exact
Weak isospin I_3	$I_3 = \pm 1/2$ from jE_7 -projector (S_u/S_d separation)	347 [B]	Theorem B; grade parity $\text{Cl}(6)$
Gell-Mann–Nishijima	$Q = I_3 + Y/2$ fully algebraic from Galois structure	347 [B]	Theorem C; no free parameter
Colour charge $r/g/b$	Z_3 charge operator $N \bmod 3$ in GF(3)	346 [B]	Theorem C (346); 3 colours = 3 elements
Chirality S_u / S_d	S_u = even $\text{Cl}(6)$ grades = right-handed	349 [B]	S_d = odd = left-handed; γ_5 -eigenvalue
$SU(2)_L$ doublet	Doublet = S_d -sector (odd grades)	349 [B]	Theorem D; ν_R structurally absent
3 generations	3 Frobenius cycles of root orbits	348 [B]	Theorem A; order = 3 forced
Permutation matrix quarks $\times$ leptons	Trace bilinear form GF(27)/GF(3): $f_3 \times f_4$	348 [B]	Theorem B; zero-coupling structure algebraic
Coupling weights $1/\sqrt{2}$	$f_1 \times f_3$ bilinear form; 2 of 3 couplings active	348/353 [B]	Theorem C + Doc. 353; $\Rightarrow$ Koide [B]
Koide amplitude $d/c = \sqrt{2}$	Z_3 equidistribution $\Rightarrow d/c = \sqrt{2} \Rightarrow Q = 2/3$	353 [B]	Formerly [S] ; closed 7 Sept. 2026
8 gluons	$4 \times Z_{13}$ -orbits $\times 2 Z_2$ -parities	339 [B]	Gluon number structurally forced
EW group Z_2	$\text{Gal}(\text{GF}(9)/\text{GF}(3)) = Z_2$	336 [K]	Numerically confirmed
<i>Open / not yet closed</i>			
Generation assignment (order)	Which orbit element = Gen. 1 / 2 / 3?	346 [S]	Not derived; orbits algebraically equivalent
CKM/PMNS angles (complete)	Full mixing angles from bilinear form	348 [S]	Theorem D open; structure known
<i>Experimentally established, FFGFT derivation open</i>			
Higgs H^0	$\tilde{T} \cdot m = 1$ replaces Higgs mechanism (Doc. 019)	019 [X]	Experimentally established; FFGFT derivation open

(Corrected on 1 Oct. 2026, row Hypercharge: previously “ N_{QR} = Legendre symbol” – inserting the Legendre symbol ± 1 directly gives $Y = +1/3$ (QR) or $-7/3$ (NQR); Doc. 347 uses the NQR bit $\in \{0, 1\}$ Since 1 Oct. 2026 the assignment is open [S]: charged leptons and quarks are both NQR (Doc. 346), so Y is not fixed by the bit alone (Doc. 347 Theorem A, Doc. 190 R146).)

All SM fermion quantum numbers follow from GF(27)* without free parameters · FFGFT J. Pascher 2026